Learning from Sound: How Far Does a Small CNN Get Against Pretrained CLAP on ESC-50?

Sæþór Berg Ásgeirsson and Óliver Bergsson · Introduction to Deep Learning, Fall 2026 · Code: github.com/Bougie01/audio_final_project_tol506m

A 1.2M-parameter CNN trained from scratch reaches 0.802 test accuracy on ESC-50, about twice an MFCC baseline (0.412). A frozen pretrained CLAP model with a linear probe reaches 0.963, at 24× the size. On 23 clips we recorded ourselves, the CNN falls to 0.52, while zero-shot CLAP, never trained on ESC-50, holds at 0.83.

1. Problem and data

Question. How close does a small CNN trained from scratch get to a large pretrained audio model at recognising everyday sounds? What does each cost in size and speed, and does either survive sounds recorded with our own microphones? Small models matter for real-time and on-device use, while pretrained models promise accuracy without much labelled data.

Data. ESC-50 [1] has 2,000 five-second clips in 50 balanced classes (40 each), in five groups: animals, natural soundscapes, human non-speech, domestic and urban sounds. The clips are cut from Freesound recordings (about 600 MB) and released under CC BY-NC 3.0.

Splits and pitfalls. We use the predefined folds: folds 1–3 for training (1,200 clips), fold 4 for validation (400) and fold 5 for testing (400). The folds keep all clips cut from the same source recording together, which prevents near-identical audio from leaking between train and test. Every choice was made on validation, and the test fold was scored once, at the end. Because the classes are balanced, we report accuracy together with macro-F1. Our own recordings form a separate test set that no model was trained on.

2. Approach and measuring progress

We start simple and change one thing at a time. Each model has one preprocessing function (mono, peak-normalised, 5 s), used in training and in our live app alike.

- **Baselines:** majority class, and logistic regression on the mean and standard deviation of 40 MFCCs (80 values per clip), with C = 0.1 chosen on validation.
- **CNN from scratch:** a 128-band log-mel spectrogram (22.05 kHz, 128 × 216) feeds four blocks of two 3×3 convolutions with batch norm, ReLU and max-pooling (32→256 channels). Mean-pooling over frequency, mean + max pooling over time, dropout 0.3 and a linear layer follow: 1.19M parameters in total. We train with AdamW and a one-cycle learning rate (peak $2{\cdot}10^{-3}$) for 80 epochs, keeping the best-validation checkpoint.
- **Augmentation ablation:** no augmentation → SpecAugment [3] (random time shift, two frequency and two time masks) → SpecAugment + mixup [4] (α = 0.4). Each variant is trained with 3 seeds and reported as mean ± std, so a gain can be compared with seed noise.
- **Pretrained CLAP** [2] (`laion/clap-htsat-unfused`, frozen, 48 kHz input). *Zero-shot:* each clip is matched to the nearest of 50 text prompts, comparing two prompt wordings. *Linear probe:* logistic regression on the 512-d audio embeddings (C = 10, chosen on validation). *MLP head:* one hidden layer of 512 units, 3 seeds.

Every run is logged to `runs/results.csv`, and learning curves are kept per run.

Model	Params	CPU ms	Val acc	Test acc	Test F1	Own rec. (n = 23)
Majority class	–	–	0.020	0.020	0.001	–
MFCC + logistic regression	4k	–	0.450	0.412	0.390	–
CNN, no augmentation	1.19M		0.829 ± 0.010	–	–	–
CNN + SpecAugment	1.19M		0.838 ± 0.003	–	–	–
CNN + SpecAugment + mixup	**1.19M**	**63**	**0.863 ± 0.007**	**0.802 ± 0.005**	**0.788 ± 0.009**	**0.522**
CLAP zero-shot, “the sound of a X”	28.2M*		0.860	–	–	–
CLAP zero-shot, “This is a sound of X.”	**28.2M***	**429**	**0.903**	**0.893**	**0.886**	**0.826**
CLAP + linear probe	**28.2M**	**429**	**0.983**	**0.963**	**0.961**	**0.783**
CLAP + MLP head	28.5M		0.976 ± 0.001	0.961 ± 0.001	0.959 ± 0.002	–

Table 1. Results on ESC-50 (validation = fold 4, test = fold 5) and on our own recordings. ± = std over 3 seeds; – = not tested, because only the variants chosen on validation were scored on test. F1 = macro-F1. Params = weights used at inference; *the text encoder (125M more) is only needed once, to embed the 50 prompts. CPU ms = time per 5-s clip on a laptop CPU, feature extraction included.

3. Results

- **The CNN doubles the baseline:** 0.802 vs 0.412 test accuracy.
- **Mixup helps by more than the noise.** It adds 3.4 validation points over no augmentation, and all three mixup seeds (≥ 0.855) beat all three unaugmented seeds (≤ 0.835). SpecAugment alone adds 0.9 points, within the seed spread. Figure 1 shows why: without augmentation the CNN memorises the training set by epoch 42 and stalls at 0.82, while with mixup validation accuracy is still rising at epoch 80.
- **CLAP wins with little or no training.** Zero-shot CLAP beats our trained CNN (0.893 vs 0.802), and the prompt

wording alone moves it by 4 points. An MLP head is no better than a linear probe, so the embeddings are already close to linearly separable.

- **Accuracy has a cost:** CLAP's audio encoder is 24× larger and 7× slower on a CPU than the CNN.
- **Every model drops from validation to test** (CNN −6.1, MFCC −3.8, CLAP probe −2.0 points). Fold 5 seems harder, and the CNN's validation score is slightly optimistic because validation chose its epoch and augmentation.

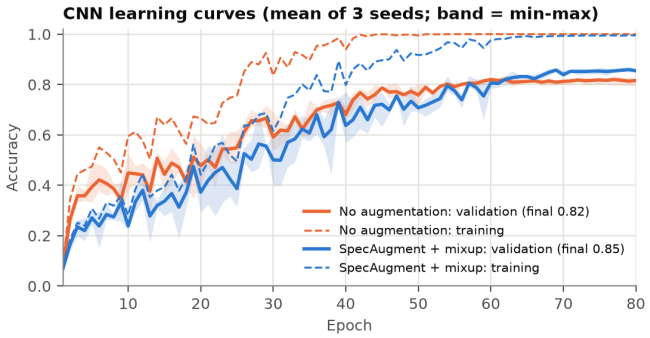


Figure 1. CNN training (dashed) and validation (solid) accuracy, mean of 3 seeds. Without augmentation the model overfits early; mixup keeps validation accuracy improving.

4. Error analysis

- **CNN (77 of 400 test clips wrong).** The weakest classes are airplane (1/8 correct), hen (2/8) and helicopter, pig and water drops (3/8). The most frequent confusions are airplane → thunderstorm, fireworks → crackling fire (4 each), wind → siren, clapping → rain and hen → laughing (3 each). Urban sounds are its weakest group (0.70, vs 0.81–0.85).
- **Why.** Engines, wind and aircraft are broadband noise with little time–frequency structure for small filters to latch onto. With 24 training clips per class, the CNN also learns texture rather than source, so dense clicks (clapping, rain) and crackles (fireworks, fire) get mixed up.
- **CLAP probe (15 errors)** still finds helicopter hardest (3/8, taken for airplane or washing machine). **Zero-shot (43 errors)** fails where the label is vague: insects (1/8), wind (2/8), hen → rooster (5 times). The probe gets all three of those classes right, so the text side is the weak point there, not the audio.

5. Real-world test: our own recordings

We recorded 23 clips of 13 household classes (e.g. coughing, door knock, keyboard typing, pouring water) with laptop microphones in ordinary rooms, through our app. All three models drop (Table 1, last column), and the more a model was fitted to ESC-50, the more it drops: the CNN loses 28.5 points, the probe 18.0 and zero-shot CLAP only 6.6. The ranking flips: zero-shot CLAP, never trained on ESC-50, becomes the best model. Our reading is that fitting to 1,200 ESC-50 clips also fits ESC-50's recording style.

The app recorded at 16 kHz. As a control, we downsampled the official test fold to 16 kHz: this costs only 1.2–2.8 points, so the sample rate explains little of the gap. The CNN misses pouring water, footsteps, snoring and wind, which CLAP gets right. For three of the five coughs, all three models agree on the same wrong label, which points to the clips (the cough may fall outside the centre 5-s window that is analysed) rather than the models. The CNN's confidence is lower on its errors (0.52 on average) than on correct answers (0.77). With 23 clips, one clip is worth 4.3 points, so these results are indicative rather than precise.

6. Limitations and next steps

- **Possible overlap:** CLAP's pretraining data includes Freesound audio, the same source as ESC-50, so its scores may be optimistic.
- **One small test fold:** with 8 clips per class, one clip moves a class by 12.5 points. ESC-50's usual protocol is 5-fold cross-validation; we kept a fixed split to have one untouched test set.
- **Closed set:** the models always pick one of the 50 classes, so speech or music gets a confident wrong label.
- **Next steps:** (1) distil CLAP into the small CNN; (2) fine-tune on our own recordings, with recording sessions split between training and testing; (3) add an "unknown" answer, using a confidence threshold calibrated on sounds outside the 50 classes; (4) run 5-fold cross-validation.

References

1. K. J. Piczak. ESC: Dataset for Environmental Sound Classification. ACM Multimedia, 2015. github.com/karolpiczak/ESC-50
2. Y. Wu, K. Chen, T. Zhang et al. Large-scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation. ICASSP, 2023.
3. D. S. Park et al. SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition. Interspeech, 2019.
4. H. Zhang, M. Cisse, Y. N. Dauphin, D. Lopez-Paz. mixup: Beyond Empirical Risk Minimization. ICLR, 2018.

AI disclosure. We used generative AI (Claude by Anthropic, via Claude Code) as a coding and writing assistant: for the user interface of our app, for writing and debugging the training and evaluation code, and for formatting the report. An AI-drafted first version of the report was rewritten in our own words. We reviewed all code and results and take responsibility for the content.